Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2022-II-Q03] | Part A (Q1-Q5)

A right square pyramid has base edge 6 and volume 54. All five vertices lie on one sphere. If its radius is reduced m/n, compute $m+n$.

正四棱锥的底边长为 6、体积为 54。五个顶点共球，球半径为最简分数 m/n。求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2026-I-Q03] | Part A (Q1-Q5)

A radius-200 solid hemisphere rests on its equatorial disk. A radius-42 sphere rests on that disk at a point P and must fit inside the hemisphere. If the area proportion of allowed P is p/q in lowest terms, find $p+q$.

半径 200 的实心上半球以赤道圆盘为底。一半径 42 的球在圆盘点 P 处与底面相切并须完全位于半球内。允许的 P 的面积占圆盘面积之比为最简分数 p/q，求 $p+q$。

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Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2020-I-Q06] | Part B (Q6-Q10)

Two holes of radii 1 and 2 are cut in a plane board, their centers 7 apart. Identical spheres rest in the holes on the same side and touch each other. For their common radius r, write $r^2 = m/n$ reduced and find $m+n$.

平面板上两圆孔半径为 $1, 2$，孔心距为 7。两个等半径球在板的同侧分别置于圆孔中，且互相相切。若球半径平方约分为 m/n，求 $m+n$。

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Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2020-II-Q07] | Part B (Q6-Q10)

Two congruent solid cones have radius 3 and height 8. Their axes meet perpendicularly at an interior point O that is 3 from each base plane. A sphere lies in their intersection. If the greatest possible radius squared is m/n reduced, find $m+n$.

两个全等实心圆锥的底面半径为 3、高为 8，轴线在内部点 O 垂直相交，且 O 距两底面均为 3。球同时包含在两个圆锥内。若最大球半径平方约分为 m/n，求 $m+n$。

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Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2021-II-Q10] | Part B (Q6-Q10)

Three spheres of radii $36, 36, 13$ are pairwise externally tangent and each tangent to two intersecting planes $\mathcal{P}, \mathcal{Q}$. Let their intersection line be ℓ. The radius-13 sphere touches $\mathcal{P}$ at U. If $\operatorname{dist}(U, \ell) = m/n$ reduced, find $m + n$.

半径为 $36, 36, 13$ 的三个球两两外切，且每球均与相交平面 $\mathcal{P}, \mathcal{Q}$ 相切。设交线为 ℓ，小球与 $\mathcal{P}$ 切于 U。若 U 到 ℓ 的距离约分为 m/n，求 $m + n$。

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Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2022-I-Q10] | Part B (Q6-Q10)

Spheres of radii $11, 13, 19$ are externally tangent in pairs. A plane cuts them in equal-radius circles with centers A, B, C corresponding to those radii. All sphere centers are on one side of the plane. Given $AB^2 = 560$, calculate AC^2.

半径分别为 $11, 13, 19$ 的三个球两两外切，同一平面截出三个等半径圆，其圆心依次为 A, B, C。三个球心位于平面同侧。已知 $AB^2 = 560$，求 AC^2。

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Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2024-II-Q08] | Part B (Q6-Q10)

A torus is generated by rotating a radius-3 circle about a coplanar axis 6 units from its center. It touches a radius-11 sphere along a circle, once while inside and once while outside the sphere. If the two contact-circle radii satisfy $r_i - r_o = m/n$ in lowest terms, find $m+n$.

将半径为 3 的圆绕其平面内距圆心 6 的轴旋转得到圆环面。该环面分别从半径 11 球的内部、外部沿圆周相切，相切圆半径为 r_i, r_o。若 $r_i - r_o = m/n$ 为最简分数，求 $m+n$。

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Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2000-II-Q12] | Part C (Q11-Q15)

Points A, B, C lie on a radius-20 sphere centered at O, with $AB = 13, BC = 14, CA = 15$. The distance from O to the filled triangle ABC is $m\sqrt{n}/k$, with n squarefree and $\gcd(m, k) = 1$. Find $m + n + k$.

半径 20、球心 O 的球面上三点满足 $AB = 13, BC = 14, CA = 15$。$O$ 到实心三角形 ABC 的距离为 $m\sqrt{n}/k$，n 无平方因子且 m, k 互质，求 $m + n + k$。

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Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2001-I-Q12] | Part C (Q11-Q15)

Find the inradius r of the tetrahedron with vertices $(6,0,0),(0,4,0),(0,0,2),(0,0,0)$. If $r = m/n$ reduced, find $m+n$.

四面体顶点为 $(6,0,0),(0,4,0),(0,0,2),(0,0,0)$，内切球半径 r 约分为 m/n，求 $m+n$。

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Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2011-I-Q13] | Part C (Q11-Q15)

A cube of side 10 is entirely above a plane. Its nearest vertex A has three adjacent vertices at heights $10, 11, 12$. If the height of A is $(r - \sqrt{s})/t$ for positive integers r, s, t, find $r + s + t$.

边长十的立方体悬在平面上方，最近顶点 A 的三个相邻顶点高度分别为 $10, 11, 12$。若 A 的高度为 $(r - \sqrt{s})/t$，r, s, t 为正整数，求 $r + s + t$。

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Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2016-II-Q14] | Part C (Q11-Q15)

Equilateral triangle ABC has side600. Points P, Q lie on opposite sides of its plane, each equidistant from A, B, C. The half-planes PAB, QAB form a 120° dihedral angle. A sphere centered at O passes through A, B, C, P, Q with radius d. Find d.

等边三角形 ABC 边长600。点 P, Q 位于其平面两侧，分别到 A, B, C 等距，半平面 PAB, QAB 所成二面角为 120°。以 O 为心的球经过 A, B, C, P, Q，半径为 d。求 d。

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Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2026-I-Q12] | Part C (Q11-Q15)

In a plane, right triangle ABC has $AB = 6, AC = 4, A = 90^\circ$. Reflect its centroid across BC to D. Spheres above the plane, tangent at A, B, C, D, have radii $1, 2, 3, r$. A distinct second plane is tangent to all four, with their centers on one side. If $r = m/n$ in lowest terms, find $m + n$.

平面内直角三角形 ABC 满足 $AB = 6, AC = 4, \angle A = 90^\circ$。重心关于 BC 对称得 D。四球位于该平面同侧并分别在 A, B, C, D 相切，半径为 $1, 2, 3, r$，且均在另一不同公切平面的同侧。将 r 化为最简分数 m/n，求 $m + n$。

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Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2026-II-Q12] | Part C (Q11-Q15)

A tetrahedron has two faces with side lengths $5\sqrt{10}, 5\sqrt{10}, 10$ and two with $5\sqrt{10}, 5\sqrt{10}, 18$. Its circumsphere and insphere centers are S, R. Write $RS = m/n$ in lowest terms and find $m + n$.

四面体两面边长为 $5\sqrt{10}, 5\sqrt{10}, 10$，另两面为 $5\sqrt{10}, 5\sqrt{10}, 18$。外接球、内切球圆心为 S, R。将 RS 化为最简分数 m/n，求 $m + n$。

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